Step 2a - H4 - Microwave manufacturer design simulation

Market scan

Market size

The buyer pool is the industrial microwave heating equipment industry itself: ~$1.4-1.6B/yr equipment sales (2025) carried by a few dozen established OEMs plus a long tail of regional/tunnel-machine builders. The comparable software market — electromagnetic simulation software overall — is ~$1.5-2.9B/yr growing 10-18%/yr, but the microwave-heating slice of that is tiny (order ~$30-80M/yr), because today it is served as a byproduct of general EM tools. A purpose-built applicator/cavity design tool addresses this niche plus the currently unmonetized 'in-house code + trial-and-error' segment.

Region	Size ($B)	Region	Size ($B)
Europe	~$0.4-0.5B (densest OEM cluster: Germany/ France/Italy/UK; European leaders ~35-40% of global microwave-dryer share)	Asia	~$0.5-0.6B (largest regional equipment market ~35%; many mid/low-tier machine builders in China/Japan/India)
America	~$0.4-0.5B (~25-30%; US specialists + component/generator vendors)	Africa	~$0.05-0.15B (incl. Middle East; small OEM presence, mostly import market)

Target customers

A concentrated, easy-to-map customer set: roughly 15-25 established industrial microwave OEMs (Muegge, Püschner, Sairem, Linn High Therm, Industrial Microwave Systems, Ferrite/Microwave Techniques, AMTek, Thermex, Cellencor, plus generator/component vendors like Richardson, CPI, e2v, Hitachi, Toshiba), ~30-80 mid-size specialists, and a long tail of ~100-300 smaller/regional tunnel-equipment builders concentrated in Asia. They are reachable through a handful of channels (AMPERE, IMPI conferences, trade fairs), and each has an engineering department that already understands the problem — the best-defined customer group of the three hypotheses, but also the smallest.

Customer size	Approx. number	Market share (in %)
Large	~10-20 OEMs	~55-70% of equipment revenue
Medium	~30-80 specialist OEMs	~25-30%
Small	~100-300 small/regional builders	~5-15%

Players offering a solution today

The design-simulation market is dominated by general-purpose EM suites: Ansys HFSS, Dassault CST Studio Suite, Altair FEKO, COMSOL Multiphysics+RF together take ~70-80% of EM-simulation spend (top-5 CAE vendors hold ~2/3 of the ~$10B CAE market). These tools were built for telecom/electronics, so microwave-heating workflows (temperature-dependent dielectrics, moving loads, thermal runaway, calorimetric power balance) must be rebuilt by each OEM's engineers. Only one small vendor (QWED, Poland, QuickWave) sells heating-dedicated modules; the rest of the niche is in-house codes, consultancies and academia — the market is effectively one strong incumbent category plus one proven niche precedent.

Player	Type of solution	Market share (in %)

Large (Ansys/Synopsys HFSS, Dassault SIMULIA CST, Altair FEKO/HyperWorks, COMSOL RF, Keysight)	general 3D EM solvers with optional thermal coupling; pricing ~$10k (COMSOL) to ~$50k+ (HFSS) per seat	~70-80% of EM-sim spend
Medium (QWED QuickWave, Remcom XFdtd, IMST Empire)	specialist FDTD tools; only QuickWave offers a dedicated microwave-heating module	~10-15%
Small (open-source openEMS/MEEP, in-house OEM codes, MW design consultancies, university groups)	free/expert tools and bespoke simulation services filling gaps the suites leave	~10-15%

Regulation

Regulation	Jurisdiction	What it means for us
IEC/EN 60519-6	International/EU	Safety requirements for industrial microwave heating installations (leakage, interlocks, emission limits 800 MHz-6 GHz ISM); design-stage simulation of leakage and hot spots directly supports conformity
CISPR 11 / EN 55011 (Group 2 ISM equipment)	International/EU	RF emission limits for ISM equipment; applicator/cavity design determines compliance, so simulation that predicts emissions shortens certification cycles
ITU Radio Regulations ISM bands	Global, region-dependent	Equipment must operate inside ISM allocations (915 MHz mainly Americas, 2.45/5.8 GHz worldwide, plus national variants like 896 MHz UK/922 MHz JP); multi-band-aware design tools cover export markets
Directive 2013/35/EU (worker EMF exposure)	EU	Minimum worker EMF-exposure requirements influence applicator shielding/feeder design — a simulation output OEMs must document
FCC Part 18	USA	US rules for ISM equipment emissions; required for machines exported to the largest single market

Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input
Name	Devin (agent draft)	Devin (agent draft)	Date	2026-10-30	2026-10-30	Name			Date	

Decision	Decision	Go to next step	Go to next step	Go to next step	Rework	Rework	Rework	Drop the hypothesis	Drop the hypothesis	Drop the hypothesis